Information Visualization

Digital storytelling

Lesson 7

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Table of contents

01

Narrative structures

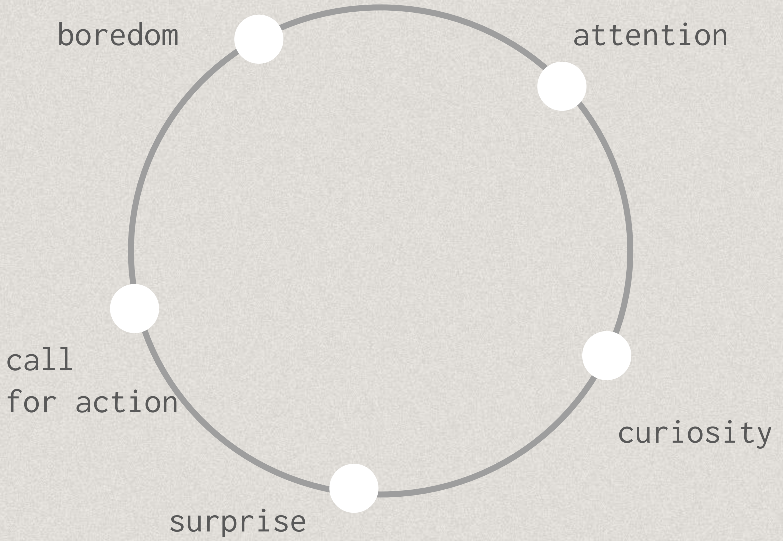
How to organise contents and drive the user in a web project based on information visualisation

The narrative

The engagement circle

- **Entertainment** usually triggered by boredom (e.g. in waiting room, cafèteria)
- Draw **attention** attract the user (e.g. give her a screen)
- Stimulate **curiosity**, provide introductory information
- Discovery by visual storytelling: learn something **unexpected** (the *so what*)
- **Recommendation:** call for action (read a book or article, listen to music, watch a movie)

Build a climax!



The content organisation

Logical blocks

Introductory texts

- **Title:** short, effective. Avoid long subtitles
- **Introduction:** background information
- **Problem statement:** what we do not know, what we want to discover
- **Motivation:** why it would be important to know it
- **Research questions:** could be a sentence or actual questions
- **Approach/methodology/sources:** which data sources have been used, whether they have been post-processed or merged, which types of analysis have been performed

Explain the audience why they should care, stimulate curiosity

Summarise what to look forward to

Gives credibility to results

The content organisation

Logical blocks

Mapping charts to questions

Each research question should be deconstructed in **sub-questions** when appropriate.

One-to-one mapping between charts and sub-questions should create the scaffolding of the story.

Among all the charts that are shown in the story, **one has to pop out**, being the one that includes the “unexpected” information. It can be a “all-in-one” chart or the result of a crescendo.

The chart is significant as a stand-alone visualisation OR it is meaningful after having seen all the previous ones OR it is meaningful when compared to sibling ones.

The content organisation

Charts sections

Must include:

- **The title of the chart.** Readers will first look at the message expressed in the title and then will look for confirmation in the chart

The content organisation

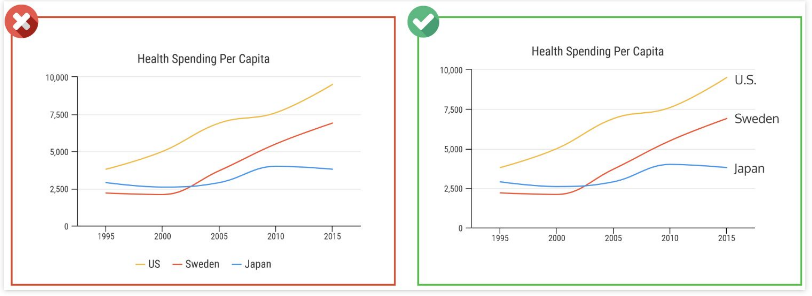
01

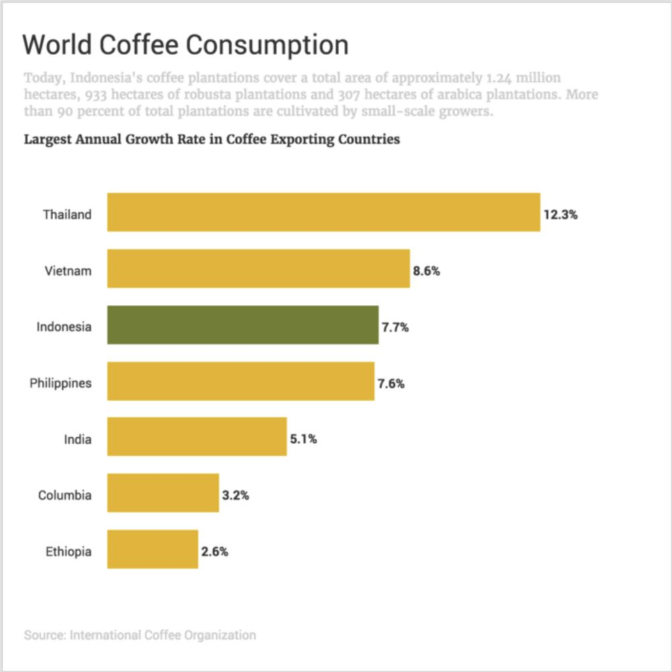
Logical blocks

Charts sections

Must include:

- **A legend (optional if trivial).** Include axes ranges, units of measure, meaningful names of data series. Do not use generic names e.g. “count”. You may also decide to label data series directly (if they are little)



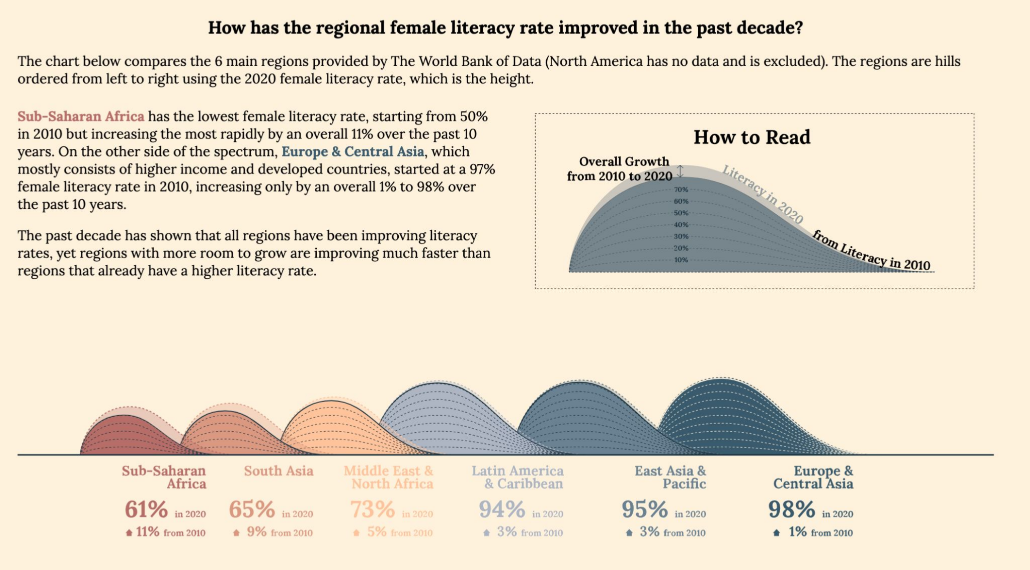


The content organisation

01

Logical blocks

Charts sections



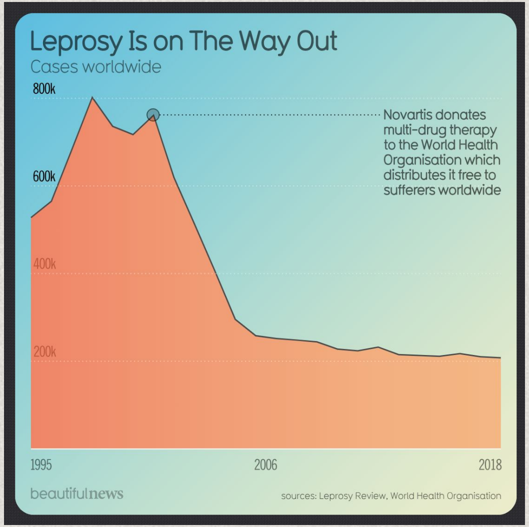
If needed, explain **how to read the chart.**

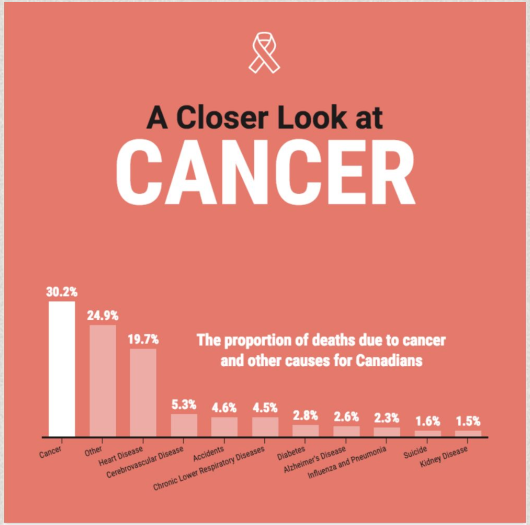
The content organisation

01

Logical blocks

Charts sections





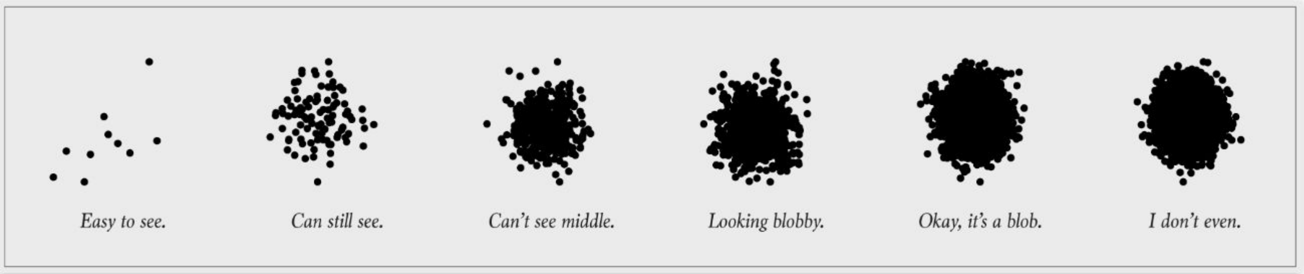
Annotate charts to focus the attention on "hotspots".

Provide additional information that explains the visual pattern (e.g. here the fast decreasing slope of the line)

The content organisation

Logical blocks

Charts sections



Find the least amount of data that confirms your pattern to make the chart easy to read.

The content organisation

Charts sections

Do not ask your reader to do the math. If you claim you are showing something, and you refer in your text to some number, that number must appear in the chart too (e.g. a percentage).

The content organisation

Charts sections

Should include:

- **An explanatory text.** Explain details of the data, guide the reader through the elements in the chart that are important to you.

The writing style

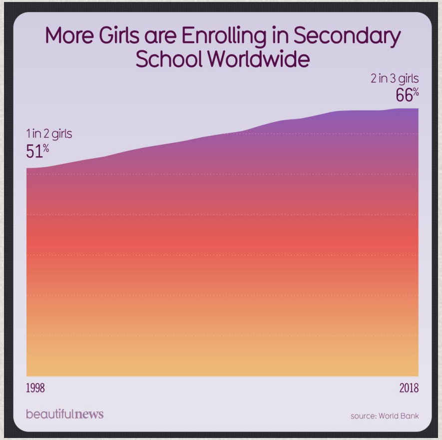
Keep it short

Impact VS explanations

Some texts must be short and **memorable**, e.g. titles.

Titles are most likely the only thing readers will read.

Titles should report the **take-home message** of a visualization. Should not be a description of what the graph shows.



The writing style

Keep it short

Impact VS explanations

Other texts need to be explanatory.

Explain **results, conclusions,** salient points, e.g. in lists.

Do not assume the **consequences** of your discourse are clear or obvious.

The writing style

Feature	Description	EDA Bias	KD Bias
Chart Count	Number of distinct charts in the doc or section	>5 charts	≤5 charts
Chart Diversity	Variety of chart types (bar, scatter, box, etc.)	≥4 types	1 type
Annotations	Explanatory notes, arrows, highlights, captions in/around charts	Little	Many
Hypothesis Language	Use of interpretive or claim-based text ("this shows...", "suggests...")	Observational	Interpretive
Iteration Evidence	Repeated views of same variables or charts with tweaks	Many tweaks	None
Polish & Finality	Layout, consistency, export quality (plot aesthetics, colors, labels)	Raw	Finalized
Source & Reproducibility	Mention of dataset, toolchain, or repo/code	Unclear	Transparent

Interpretive language

EDA VS KD

When you perform EDA you write in a different way than when you present results of a Knowledge Discovery journey.

Never ever...

02

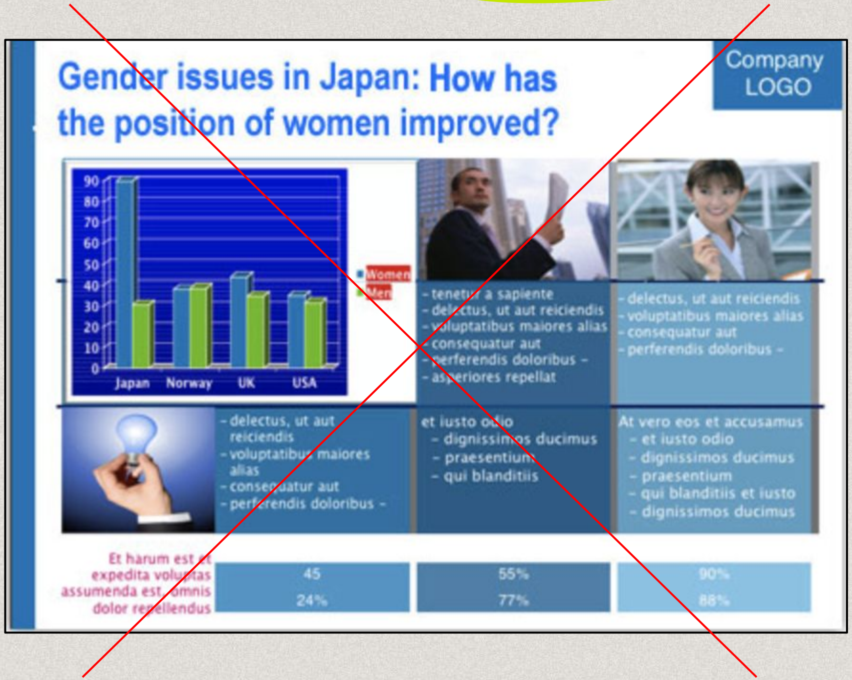
Nothing given for granted

SLIDUMENT

Avoid **imbalances** in the presentation.

Do not put too much text as you would do in a document (remember the F-pattern)

Do not be too short with catchy phrases only (you still need to explain things!)



The content organisation

01

Logical blocks

Conclusions

A story can be long, with many important focus points that all together must lead to some conclusions. Conclusions must be **stated explicitly at the end of the data story.**

Summarise all the main outcomes you have reached (go through the whole story again), provide your overall interpretation of results, and, if applicable, **recommend actions.**

The content organisation

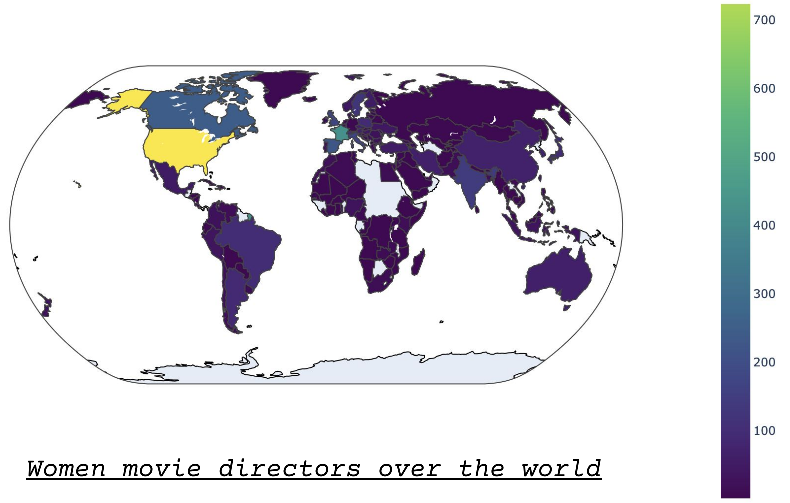
01

Logical blocks

Identify the tension

When recommending actions you must think in **dialectical terms**. There is a **tension**, highlighted by data, and a resolution, driven by your **interpretation** of data, which leads you to define a **solution**.

E.g. US has the highest number of female directors. Factors such as cultural barriers, lack of funding, and limited access to filmmaking education likely contribute to the low representation of female directors in regions like Africa and Asia. The story sheds light on systemic challenges and opportunities for greater gender equality in the movie industry.



Women movie directors over the world

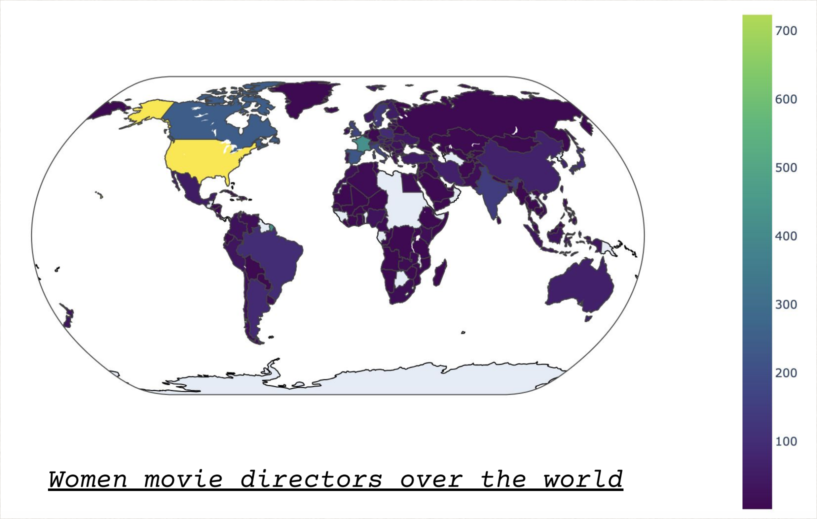
The content organisation

Logical blocks

Which actions?

Call for actions read like “After reading this story my readers should …” and include verbs like the following ones:

Accept | Agree | believe | buy | change | collaborate | contribute | create | debate | decide | defend | desire | examine | do | empower | engage | facilitate | implemente | increase | influence | learn | move | plan | promote | pursue | recommend | reconsider | reduce | reflect | respond | reuse | secure | share | simplify | validate | verify …



Women movie directors over the world

What could be a call for action here?

02

Communication

Leverage visual aspects and phenomena typical of the web in the communication of your results

The page

02

Eye focus

F pattern

Short sentences in short paragraphs.

Make strategic interruptions in the flow of text with visuals, titles, actions.

Eyetracking by Nielsen Norman Group nngroup.com NN/g

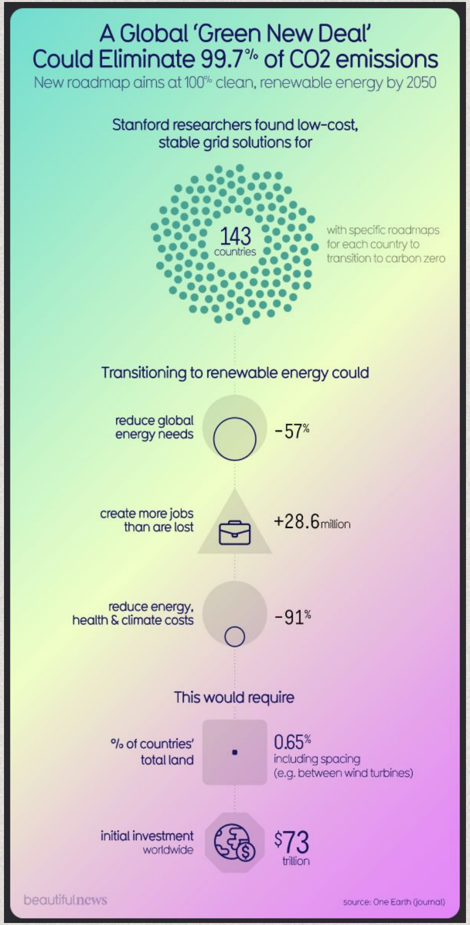
Colors and space

Negative space

Use **white spaces** to help the eye to fucs.

Create **symmetries** and highlight the flow.

Use the **minimum number of colors** possible. Repeat the colors through the story to show the way.



Scrollytelling

Linear reading

Scrolling is the most common action on the web. It fosters linear reading of a story, but it can trigger **new behaviours** other than just moving on on the page.

The animation of every new section coming in the screen **guides the attention** of the user on specific moments of the story, and sets the pace of the story (like in a melody).

Example The tale of two pandemics: COVID-19, Auckland and the Delta variant

Scrollytelling

Use with moderation!

It creates **addiction** and too many animations/contents may be overwhelming, e.g. like in social media platforms.

Try to foster **slow looking activities** that encourage critical thinking instead, while scrolling.

Use scrollytelling only as a **complementary solution**, e.g. when a complex chart requires step-by-step explanations or introductions, guiding the reader in what to focus on iteratively.

Animation

Subtleties

Animation in UX must be unobtrusive, brief, and subtle.

We can use animation to:

- give noticeable **feedback** that a user action has been recognized by the system, e.g. click on a hamburger menu opens a sliding menu
- **State-change**, e.g. when clicking on a pencil icon it changes shape/color/content to show a different modality (edit modality on)
- navigation **metaphors**, e.g. zooming to show hierarchical content or sliding screens to show progress in a journey.
- to enhance **signifiers**, e.g. swiping left on a list element shows buttons, swiping down a content sends you back.
- show **iteratively** elements of a section in the sequence in which they should be read.

Animation

Use with moderation!

It should be used to support the linear reading of the story, rather than as a mere aesthetic feature.

We are sensitive and prone to be distracted by any type of motion (meaningful or not). That's why animation can easily become annoying: it's hard to stop attending to it, and, if irrelevant to the task at hand, it can substantially degrade the user experience (as any web user who has encountered a moving advertisement can attest). Source

Animation

01

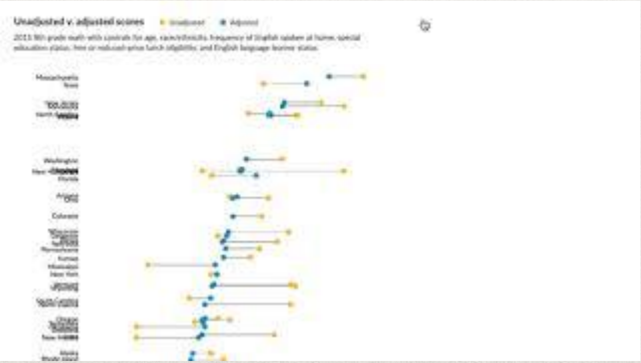
Interaction

In information visualisations

Can be used (wisely) for several purposes:

Update data when setting different parameters, e.g. applying a filter, or a different feature.

Notice that the view (the chart) is the same, while the data series change (values on the x axis), and with them also values on the y axis (some disappear and others come in).



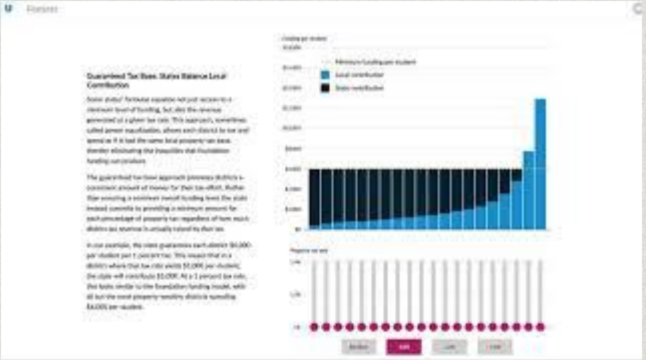
Animation

In information visualisations

Can be used (wisely) for several purposes:

To **change data points** while scrolling, so as to follow a narrative (e.g. explanatory text in the left column).

Again, notice the view (the chart) and all the coordinates (values on x axis) are is still the same, while the values data series change (y axis).



Animation

01

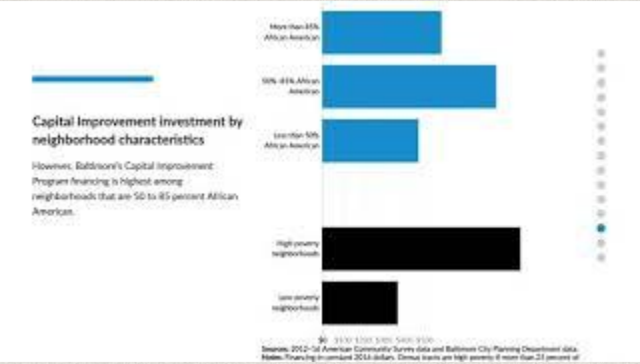
Interaction

In information visualisations

Can be used (wisely) for several purposes:

To **transition** between one chart and another one.

In this case, the view changes while the data are shown in a different way (e.g. different aggregation, filtered values).



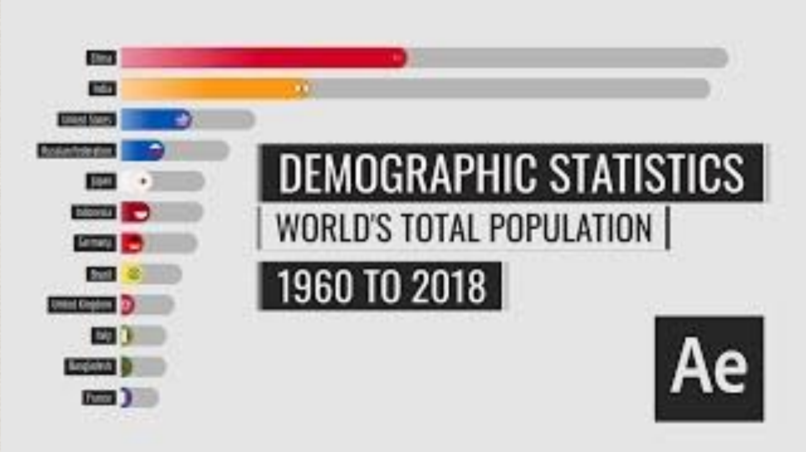
Animation

In information visualisations

Can be used (wisely) for several purposes:

To **show a trend.**

Animation can be used to populate a chart iteratively and to show differences between (time) series.



Animation

In information visualisations

Can be used (wisely) for several purposes:

To **highlight patterns.**

Animation can be used to focus the attention of the reader on a data sample

Interactivity

Not just animation

Interactive charts allow users to engage, and manipulate the view on demand. They use animations as a way to give feedback to readers.

They are eye-pleasing, engaging, and may **reveal patterns** that are not visible in static charts.

They are good for **large datasets**, where to show several views and where more questions could be investigated.

Interactivity

Not just animation

It can be used for several purposes:

Filter out data from a view that includes several data series, so that you can focus on trends and patterns of samples.

See some examples

Interactivity

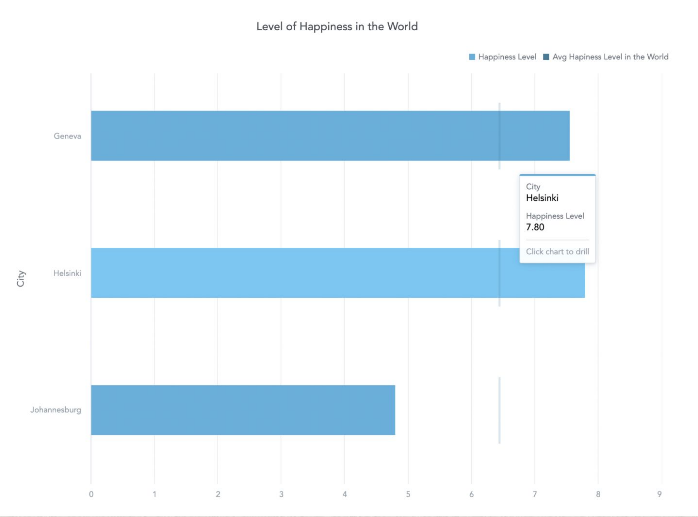
01

Interaction

Not just animation

It can be used for several purposes:

Drill down. Show detail of aggregations, e.g. labels and additional information.



02

Hands-on

A trivia! With the proper <u>soundtrack</u>

Trivia time

02

Get amazing prizes

Go to the form

1. Go to the google form https://forms.gle/aLdW2HnXzjQXgq9F6
2. I'll show you some numbered **slides**, you will have to answer a question in the form labeled with the same number.
3. Each question is **timed**. Be quick!

Thanks!

Do you have any questions?

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https://github.com/marilenadaquino/information_visualization